MSG No. 188.220 $P_c\bar{6}c2$ [Type IV, hexagonal]

Table 1: Wyckoff site: 2a, site symmetry: -6'm'2

No.	position	mapping
1	$[0,\ 0,\ 0]$	[1,2,3,4,5,6,19,20,21,22,23,24]
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	[7,8,9,10,11,12,13,14,15,16,17,18]

Table 2: Wyckoff site: 2b, site symmetry: -6m'2'

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	[1,2,3,7,8,9,16,17,18,22,23,24]
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	[4,5,6,10,11,12,13,14,15,19,20,21]

Table 3: Wyckoff site: 2c, site symmetry: -6'm'2

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	[1,2,3,4,5,6,19,20,21,22,23,24]
2	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	[7,8,9,10,11,12,13,14,15,16,17,18]

Table 4: Wyckoff site: 2d, site symmetry: -6m'2'

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	[1,2,3,7,8,9,16,17,18,22,23,24]
2	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	[4,5,6,10,11,12,13,14,15,19,20,21]

Table 5: Wyckoff site: 2e, site symmetry: -6'm'2

No.	position	mapping
1	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	[1,2,3,4,5,6,19,20,21,22,23,24]
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	[7,8,9,10,11,12,13,14,15,16,17,18]

Table 6: Wyckoff site: 2f, site symmetry: -6m'2'

No.	position	mapping
1	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	[1,2,3,7,8,9,16,17,18,22,23,24]
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	[4,5,6,10,11,12,13,14,15,19,20,21]

Table 7: Wyckoff site: 4g, site symmetry: 3m'.

No.	position	mapping
1	$[0,\ 0,\ z]$	[1,2,3,22,23,24]
2	$[0,\ 0,\ -z]$	[4,5,6,19,20,21]

continued ...

Table 7

No.	position	mapping
3	$\left[0,\, 0,\, \frac{1}{2}-z\right]$	[7,8,9,16,17,18]
4	$\left[0,\, 0,\, z+\frac{1}{2}\right]$	[10,11,12,13,14,15]

Table 8: Wyckoff site: 4h, site symmetry: 3m'.

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, z\right]$	[1,2,3,22,23,24]
2	$\left[\frac{1}{3},\, \frac{2}{3},\, -z\right]$	[4,5,6,19,20,21]
3	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{2}-z\right]$	[7,8,9,16,17,18]
4	$\left[\frac{1}{3},\, \frac{2}{3},\, z+\frac{1}{2}\right]$	[10,11,12,13,14,15]

Table 9: Wyckoff site: 4i, site symmetry: 3m'.

No.	position	mapping
1	$\left[\frac{2}{3},\, \frac{1}{3},\, z\right]$	[1,2,3,22,23,24]
2	$\left[\frac{2}{3},\, \frac{1}{3},\, -z\right]$	[4,5,6,19,20,21]
3	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{2}-z\right]$	[7,8,9,16,17,18]
4	$\left[\frac{2}{3},\, \frac{1}{3},\, z+\frac{1}{2}\right]$	[10,11,12,13,14,15]

Table 10: Wyckoff site: 6j, site symmetry: m'm'2

No.	position	mapping
1	$[x,\, -x,\, 0]$	[1,6,20,23]
2	$[x,\, 2x,\, 0]$	[2,4,21,24]
3	$[-2x,\, -x,\, 0]$	[3,5,19,22]
4	$\left[-2x,\, -x,\, \frac{1}{2}\right]$	[7,10,15,17]
5	$\left[x,\, -x,\, \frac{1}{2}\right]$	[8,11,13,18]
6	$\left[x,\, 2x,\, \frac{1}{2}\right]$	[9,12,14,16]

Table 11: Wyckoff site: 6k, site symmetry: mm'2'

No.	position	mapping
1	$\left[x,\, -x,\, \frac{1}{4}\right]$	[1,8,18,23]
2	$\left[x,\, 2x,\, \frac{1}{4}\right]$	[2,9,16,24]
3	$\left[-2x,\, -x,\, \frac{1}{4}\right]$	[3,7,17,22]
4	$\left[x,\, 2x,\, \frac{3}{4}\right]$	[4,12,14,21]
5	$\left[-2x,\, -x,\, \frac{3}{4}\right]$	[5,10,15,19]
6	$\left[x,\, -x,\, \frac{3}{4}\right]$	[6,11,13,20]

Table 12: Wyckoff site: `12l`, site symmetry: `m'..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,20]`
2	$[-y,\ x-y,\ 0]$	`[2,21]`
3	$[-x+y,\ -x,\ 0]$	`[3,19]`
4	$[x,\ x-y,\ 0]$	`[4,24]`
5	$[-x+y,\ y,\ 0]$	`[5,22]`
6	$[-y,\ -x,\ 0]$	`[6,23]`
7	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	`[7,15]`
8	$\left[x,\ y,\ \frac{1}{2}\right]$	`[8,13]`
9	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	`[9,14]`
10	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	`[10,17]`
11	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[11,18]`
12	$\left[x,\ x-y,\ \frac{1}{2}\right]$	`[12,16]`

Table 13: Wyckoff site: `12m`, site symmetry: `m..`

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{4}\right]$	`[1,8]`
2	$\left[-y,\ x-y,\ \frac{1}{4}\right]$	`[2,9]`
3	$\left[-x+y,\ -x,\ \frac{1}{4}\right]$	`[3,7]`
4	$\left[x,\ x-y,\ \frac{3}{4}\right]$	`[4,12]`
5	$\left[-x+y,\ y,\ \frac{3}{4}\right]$	`[5,10]`
6	$\left[-y,\ -x,\ \frac{3}{4}\right]$	`[6,11]`
7	$\left[x,\ y,\ \frac{3}{4}\right]$	`[13,20]`
8	$\left[-y,\ x-y,\ \frac{3}{4}\right]$	`[14,21]`
9	$\left[-x+y,\ -x,\ \frac{3}{4}\right]$	`[15,19]`
10	$\left[x,\ x-y,\ \frac{1}{4}\right]$	`[16,24]`
11	$\left[-x+y,\ y,\ \frac{1}{4}\right]$	`[17,22]`
12	$\left[-y,\ -x,\ \frac{1}{4}\right]$	`[18,23]`

Table 14: Wyckoff site: `12n`, site symmetry: `.m'.`

No.	position	mapping
1	$[x,\ -x,\ z]$	`[1,23]`
2	$[x,\ 2x,\ z]$	`[2,24]`
3	$[-2x,\ -x,\ z]$	`[3,22]`
4	$[x,\ 2x,\ -z]$	`[4,21]`
5	$[-2x,\ -x,\ -z]$	`[5,19]`
6	$[x,\ -x,\ -z]$	`[6,20]`
7	$\left[-2x,\ -x,\ \frac{1}{2}-z\right]$	`[7,17]`
8	$\left[x,\ -x,\ \frac{1}{2}-z\right]$	`[8,18]`
9	$\left[x,\ 2x,\ \frac{1}{2}-z\right]$	`[9,16]`
10	$\left[-2x,\ -x,\ z+\frac{1}{2}\right]$	`[10,15]`
11	$\left[x,\ -x,\ z+\frac{1}{2}\right]$	`[11,13]`
12	$\left[x,\ 2x,\ z+\frac{1}{2}\right]$	`[12,14]`

Table 15: Wyckoff site: 24o, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x-y,\ z]$	[2]
3	$[-x+y,\ -x,\ z]$	[3]
4	$[x,\ x-y,\ -z]$	[4]
5	$[-x+y,\ y,\ -z]$	[5]
6	$[-y,\ -x,\ -z]$	[6]
7	$\left[-x+y,\ -x,\ \frac{1}{2}-z\right]$	[7]
8	$\left[x,\ y,\ \frac{1}{2}-z\right]$	[8]
9	$\left[-y,\ x-y,\ \frac{1}{2}-z\right]$	[9]
10	$\left[-x+y,\ y,\ z+\frac{1}{2}\right]$	[10]
11	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	[11]
12	$\left[x,\ x-y,\ z+\frac{1}{2}\right]$	[12]
13	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[13]
14	$\left[-y,\ x-y,\ z+\frac{1}{2}\right]$	[14]
15	$\left[-x+y,\ -x,\ z+\frac{1}{2}\right]$	[15]
16	$\left[x,\ x-y,\ \frac{1}{2}-z\right]$	[16]
17	$\left[-x+y,\ y,\ \frac{1}{2}-z\right]$	[17]
18	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[18]
19	$[-x+y,\ -x,\ -z]$	[19]
20	$[x,\ y,\ -z]$	[20]
21	$[-y,\ x-y,\ -z]$	[21]
22	$[-x+y,\ y,\ z]$	[22]
23	$[-y,\ -x,\ z]$	[23]
24	$[x,\ x-y,\ z]$	[24]